

04 - IRS Communication Tunnels Module

Overview

The IRS Tunnel module maintains persistent, encrypted communications with IRS internal systems for real-time data exchange including transcripts, account status, CAF authorization verification, and compliance pulls.

Tunnel Types

Tunnel	Purpose	Protocol	IRS Endpoint
Transcript Delivery	Tax transcripts via TDS	HTTPS/mTLS	irs.gov/tds
CAF Authorization	POA/TIA validation	HTTPS/mTLS	irs.gov/caf
Account Status	Real-time account inquiry	HTTPS/mTLS	irs.gov/acct
CP Notice Feed	Compliance notice delivery	SFTP/TLS	irs.gov/notices
TC Pull	Transaction code queries	HTTPS/mTLS	irs.gov/rtd

Connection Parameters

```
irs_tunnel:
  efin: "748335"
  keep_alive_interval_seconds: 60
  max_concurrent_sessions: 10
  psk_rotation_days: 30
  dead_peer_detection: true
  reconnect_on_drop: true
  reconnect_attempts: 5
  reconnect_backoff_seconds: [5, 10, 20, 40, 60]
  tls_version: "1.3"
  cipher_suite: "TLS_AES_256_GCM_SHA384"
  mutual_tls: true
  client_cert_path: "/certs/irs-client.pfx"
```

Tunnel Session Lifecycle

[Tunnel Engine Start]

- > [Load mTLS Certificate (EFIN 748335)]
 - > [Initiate Handshake with IRS Endpoint]
 - > [Session Established, Session ID logged]
 - > [Keep-Alive Pings every 60s]
 - > [Request/Response Exchange]
 - > [Session Termination or Dead Peer Detection]
 - > [Automatic Reconnection]
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Transcript Request Types

Transcript Type	IRS Code	Use Case
Tax Return Transcript	TRDBV	Verify filed return
Tax Account Transcript	TXACT	TC codes, payments, penalties
Wage and Income Transcript	WIGE	W-2/1099 cross-reference
Record of Account	RECA	Combined return and account
Verification of Non-Filing	VONF	Confirm no return filed

CAF Authorization Levels

Level	Code	Scope
Full Representation	A	All matters
Limited Representation	F	Specific tax periods
Information Only	Q	Transcripts only
Disclosure Only	D	Third-party disclosure

Security Requirements

- PSK is EFIN-specific and must not be shared across EFINS.
- mTLS client certificate must match IRS-issued certificate for EFIN 748335.
- All tunnel traffic is logged to immutable audit storage.
- PII in tunnel responses is masked in application logs (for example, SSN -> XXX-XX-XXXX).

SLA and Monitoring

- Tunnel uptime SLA target: 99.9%
- Certificate expiry alert threshold: 90 days

Metric	Alert Threshold	Critical Threshold
Tunnel Uptime	< 99.5%	< 99.0%
Session Reconnects/hour	> 3	> 10
Request Latency (p95)	> 2000ms	> 5000ms
Certificate Expiry	< 90 days	< 30 days
PSK Age	> 25 days	> 30 days